

Reviewer Pack — Zarankiewicz Bound on Disjointness Communication Complexity

Entry eb9597f7a1cd · INCONCLUSIVE

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2026-05-20 03:52:55 UTC

Zarankiewicz Bound on Disjointness Communication Complexity

Entry ID: eb9597f7a1cd **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-20 03:52:55 UTC

1. Conjecture

Field A (mathematical branch): Extremal Graph Theory **Field B** (complexity object): Communication Complexity

Statement:

For a bipartite graph G with parts of size n , the deterministic communication complexity of the disjointness problem on G is $\Omega(Z(n, 2, 2))$, where $Z(n, 2, 2)$ is the maximum number of edges in G without a $K_{\{2,2\}}$.

Rationale (proposer's reasoning):

The Zarankiewicz problem provides bounds on the edge density of graphs avoiding certain subgraphs. The disjointness problem's communication complexity is influenced by the structure of the graph, and linking it to the Zarankiewicz number could reveal structural dependencies.

Taxonomy category: LIFTING (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): f2aad48d8dd37064

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
ALGEBRIZATION	SAFE	0.95	SAFE	SAFE
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Zarankiewicz problem communication complexity disjointness - extremal graph theory communication complexity $K_{\{2,2\}}$ -free graphs - Zarankiewicz bound deterministic communication complexity lower bounds

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_bipartite_graph(n):
        A = [0] * n
        B = [0] * n
        edges = []
        for i in range(n):
            for j in range(n):
                if random.random() < 0.5:
                    edges.append((i, j))
                    A[i] += 1
                    B[j] += 1
        return A, B, edges

    def zarankiewicz_bound(n):
        return math.floor(2 * n * (n - 1) / 4)

    def disjointness_communication_complexity(A, B, edges):
        # Simplified deterministic protocol for communication complexity
        return max(max(A), max(B))

    n = random.randint(5, 40)
    A, B, edges = generate_bipartite_graph(n)

    z_bound = zarankiewicz_bound(n)
    comm_complexity = disjointness_communication_complexity(A, B, edges)

    return {
        "metric_name": "disjointness_communication_complexity",
        "metric_value": comm_complexity,
        "instances_tested": 1,
```

```

        "conjecture_holds": comm_complexity >= z_bound,
        "counterexample": "" if comm_complexity >= z_bound else f"Graph with n={n}, A={A}, B={B} h
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_dev = math.sqrt(sum((result["metric_value"] - mean_value) ** 2 for result in results) / len(results))
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_dev} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_dev} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

me': 'disjointness_communication_complexity', 'metric_value': 18, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'disjointness_communication_complexity', 'metric_value': 11, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'disjointness_communication_complexity', 'metric_value': 18, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'disjointness_communication_complexity', 'metric_value': 9, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'disjointness_communication_complexity', 'metric_value': 14, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'disjointness_communication_complexity', 'metric_value': 21, 'instances_tested': 1, 'conjecture_holds': True}

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: Multiple counterexamples show communication complexity below $Z(n, 2, 2)$. The first counterexample has $n=25$ with complexity 17, directly violating the | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	136235
2	propose	ollama_remote	qwen3:8b	0	0	206690
3	preregistration	ollama_remote	qwen3:8b	0	0	38489
4	novelty	ollama_remote	qwen3:8b	0	0	29888
5	novelty	ollama_remote	qwen3:8b	0	0	28625
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10975
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6992
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8267
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7679
10	judge	ollama_remote	qwen3:8b	0	0	31744

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 505584 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/eb9597f7a1cd.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/eb9597f7a1cd.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/eb9597f7a1cd.tar.gz` (if generated)